

Home Lab Report

Enterprise IT Environment Simulation

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May 2026

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Overview

This report documents the setup and configuration of a self-built enterprise IT home lab. The goal was to simulate a real-world IT environment and gain hands-on experience with tools used in actual IT support and administration roles — rather than relying solely on theory.

The lab covers two main areas: on-premise infrastructure using Active Directory and Group Policy, and cloud-based Microsoft 365 administration including Entra ID, Intune, Exchange Online, and SharePoint Online.

Lab Setup

Component	Details
Server	Windows Server 2022 running on a Virtual Machine
Client Machine	Dell Laptop — Windows 10 Enterprise, Domain joined
Host Machine	Personal laptop (Win 11) hosting the VM
Cloud Tenant	Microsoft 365 E3 Trial subscription.

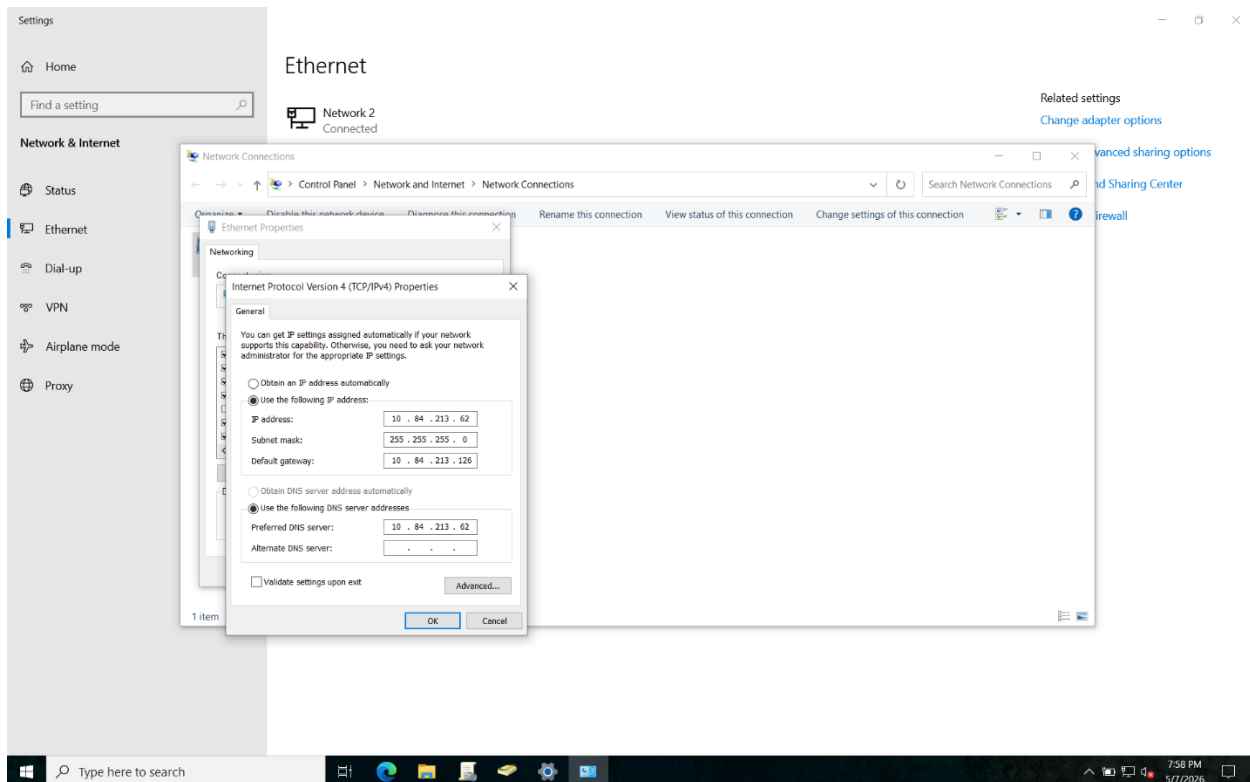
Section 1: Active Directory

This section covers the on-premise Active Directory environment built using Windows Server as a Domain Controller, connected to a Dell laptop running Windows 10 Enterprise as a domain-joined client machine.

1.1 Domain Controller Setup

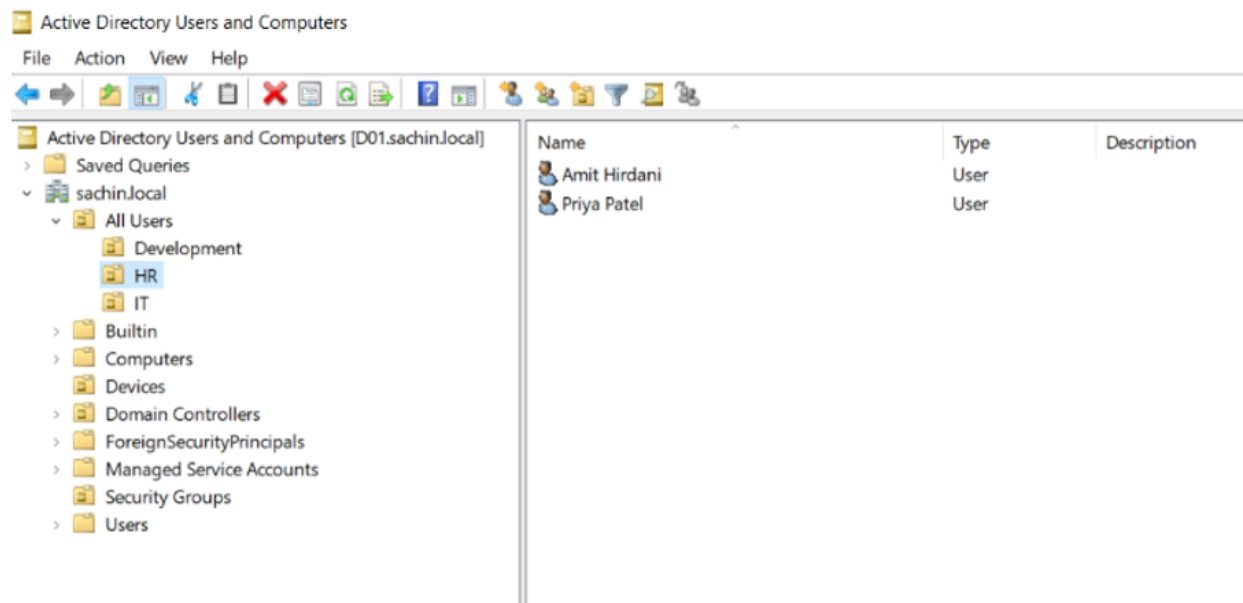
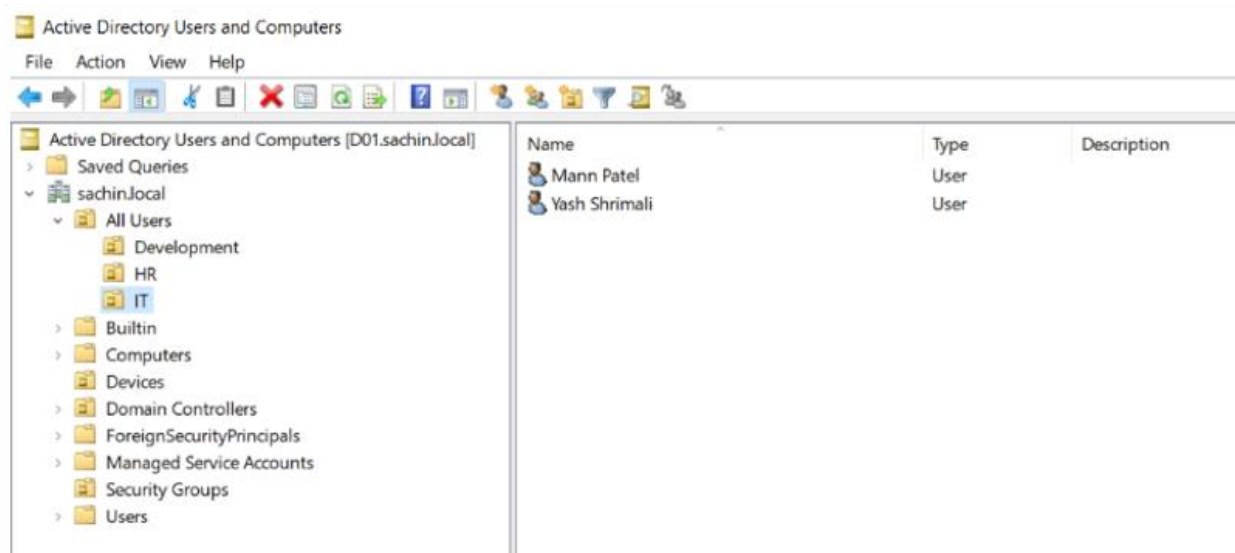
Windows Server was installed on a virtual machine and promoted to a Domain Controller by installing the Active Directory Domain Services (AD DS) role via Server Manager. A new forest and domain were created from scratch.

Static IP and DNS configuration on AD Server



1.2 Users, Computers and Organizational Units

User accounts and computer accounts were created and organized into Organizational Units (OUs) to reflect a real department structure. The Dell laptop was domain-joined and appeared under the Computers OU in Active Directory Users and Computers (ADUC).



Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers [D01.sachin.local]

Active Directory Users and Computers [D01.sachin.local] Table:

Name	Type	Description
Ayush Togadaya	User	
Sneh Patel	User	

Active Directory Users and Computers

File Action View Help

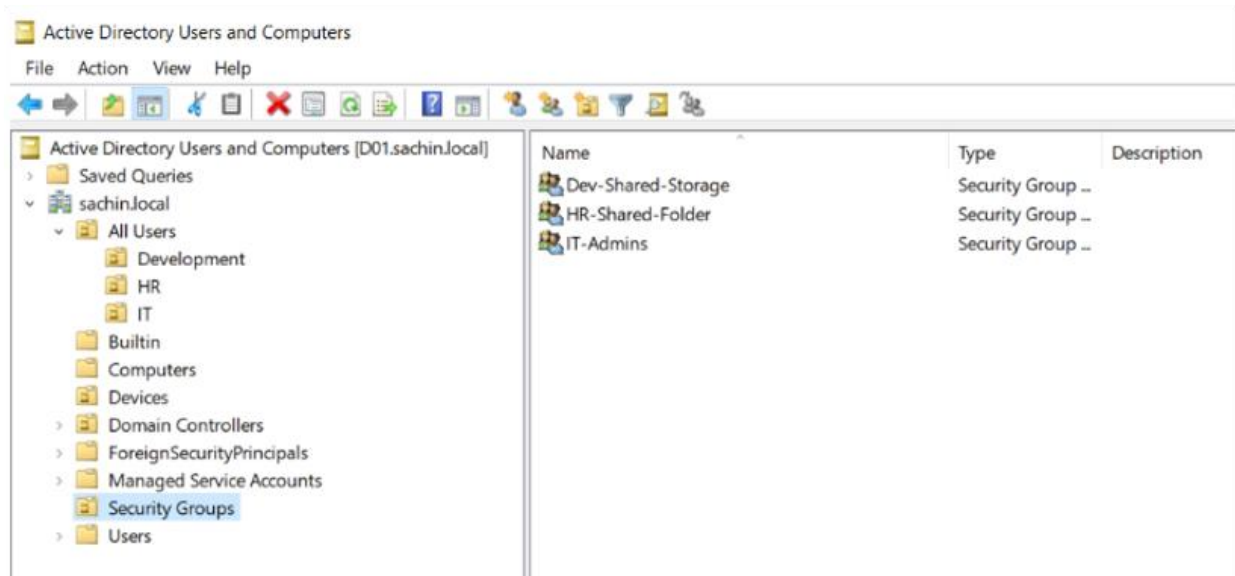
Active Directory Users and Computers [D01.sachin.local]

Active Directory Users and Computers [D01.sachin.local] Table:

Name	Type	Description
DESKTOP-2S39HU4	Computer	

1.3 Security Groups

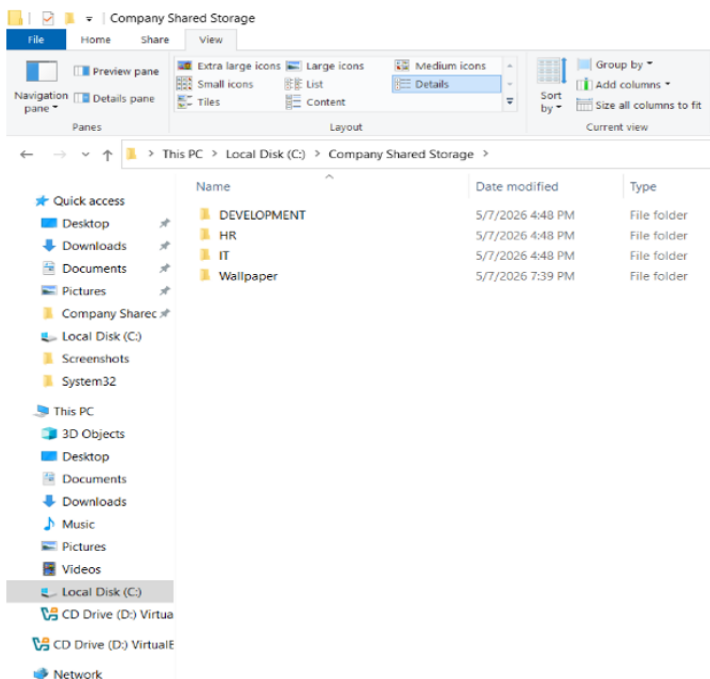
Security groups were created to manage access to resources. Users were added to groups based on their department, and permissions were assigned to the group rather than individual users — reflecting real-world best practice.



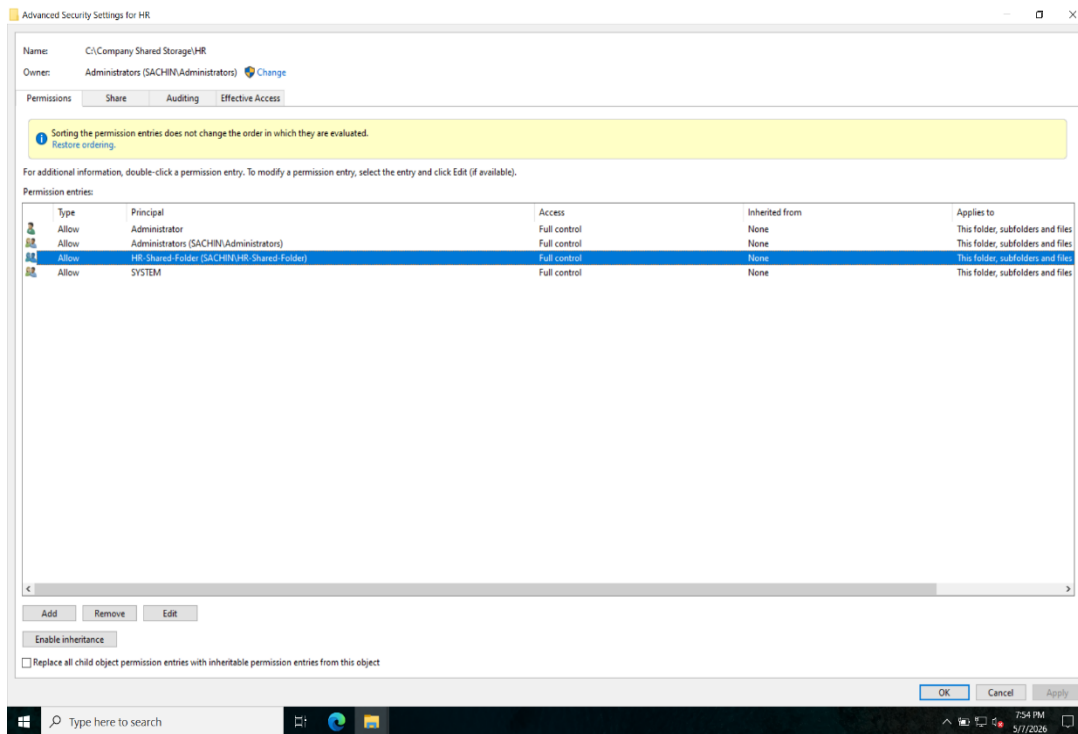
1.4 Shared Folder and NTFS Permissions

A shared folder was created on the server and made accessible to domain-joined machines. NTFS permissions were configured so that different users and groups could access only the folders they were authorized for. Users outside the permitted group could not see or access restricted folders.

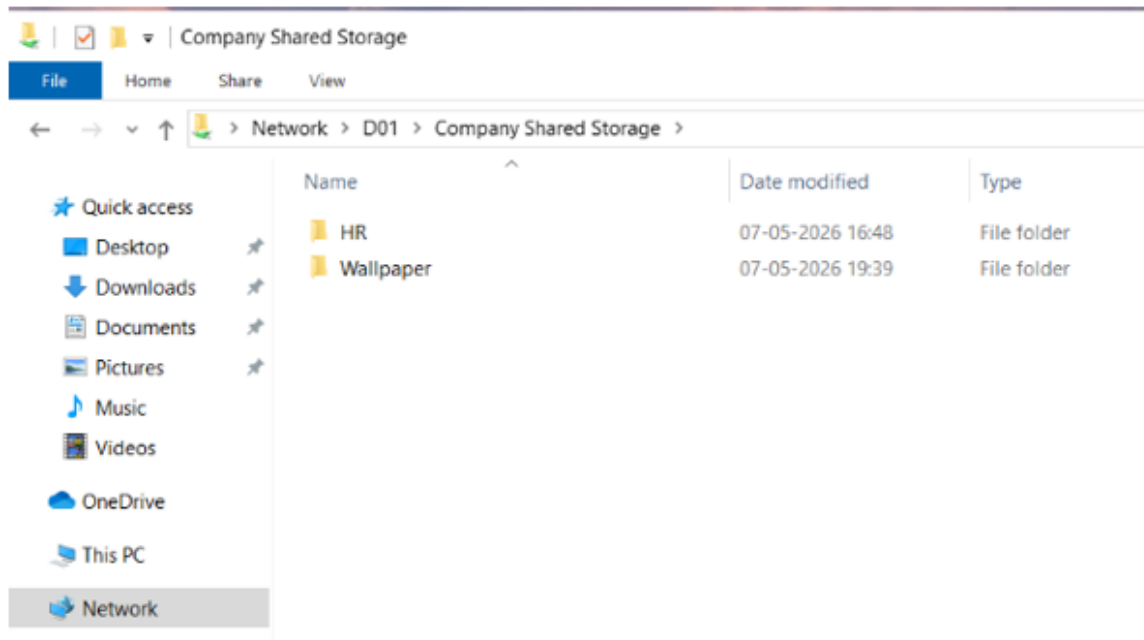
Shared folders



NTFS permission configuration



HR OU users only able to access HR folder and other shared folders.

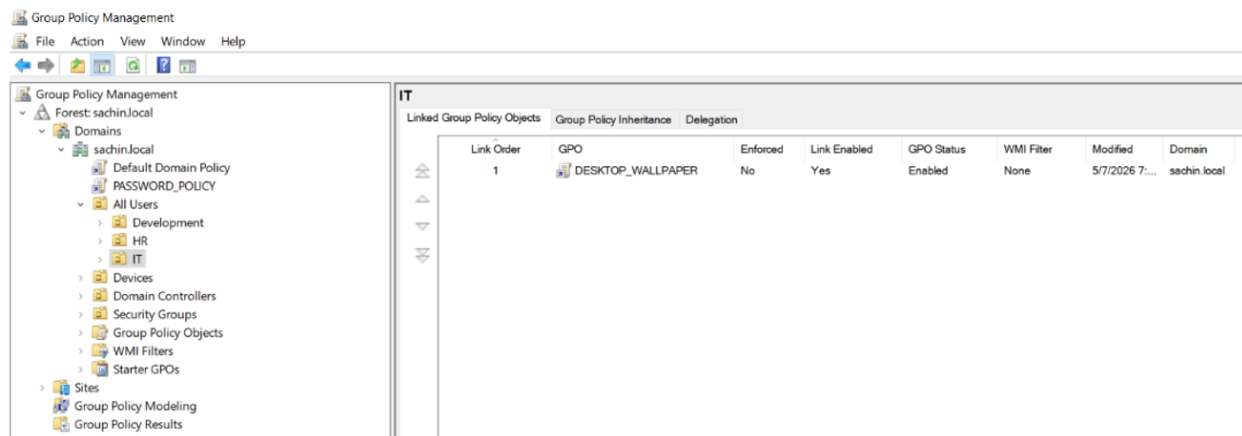


1.5 Group Policy Objects (GPOs)

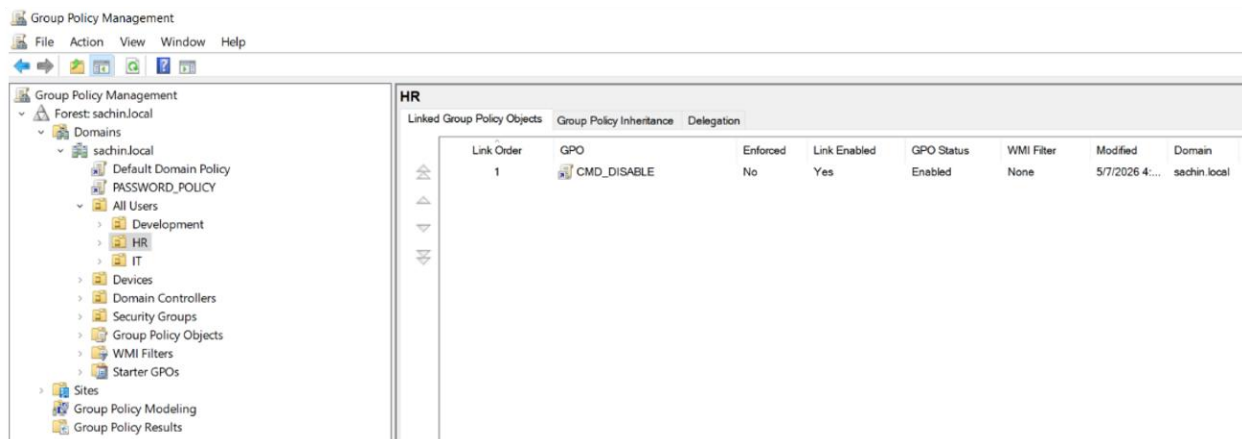
Multiple Group Policy Objects were configured and linked to the domain to enforce settings across all domain-joined machines. The following policies were implemented:

- Wallpaper enforcement — a company wallpaper was pushed to all machines via GPO
- CMD disabled — Command Prompt was restricted for standard users
- Settings app disabled — Control Panel and PC Settings were blocked for standard users
- Software deployment — 7-Zip was deployed automatically to domain machines via GPO software installation

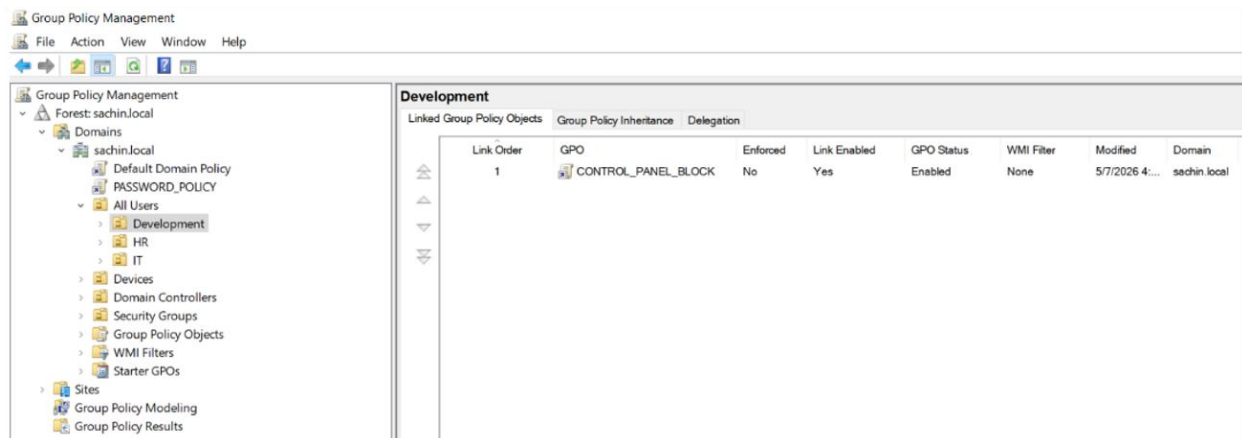
Wallpaper enforcement policy



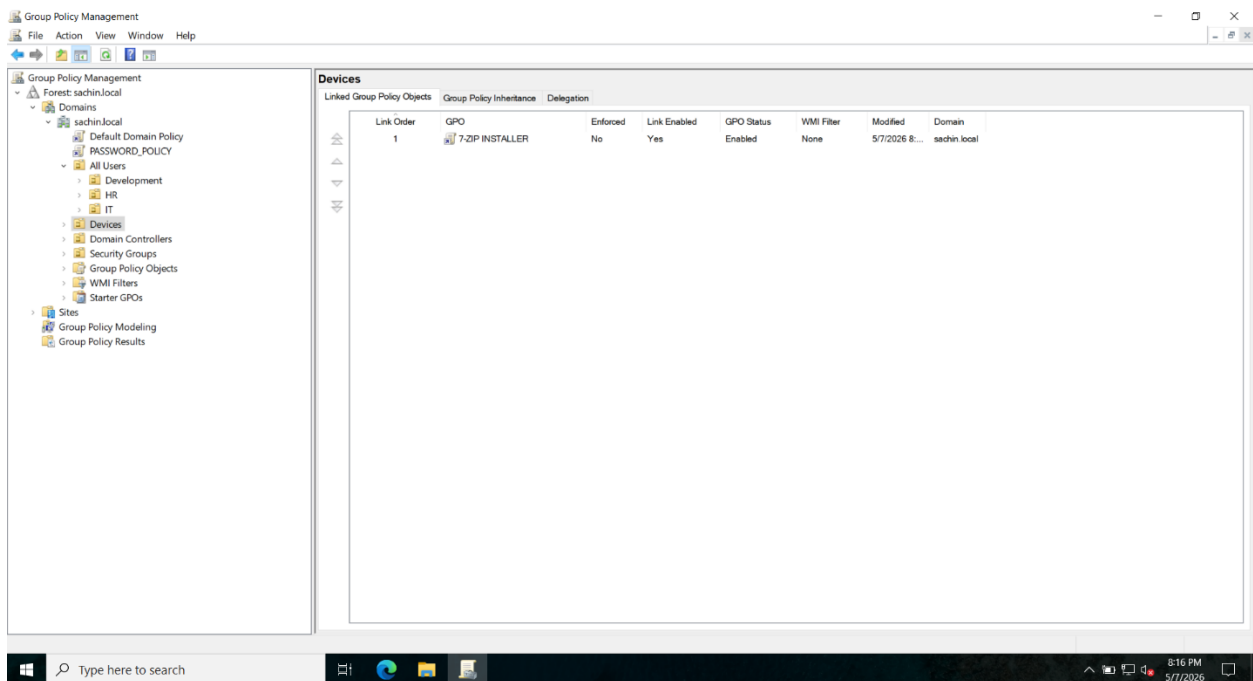
Command prompt disabled



Control Panel Blocked



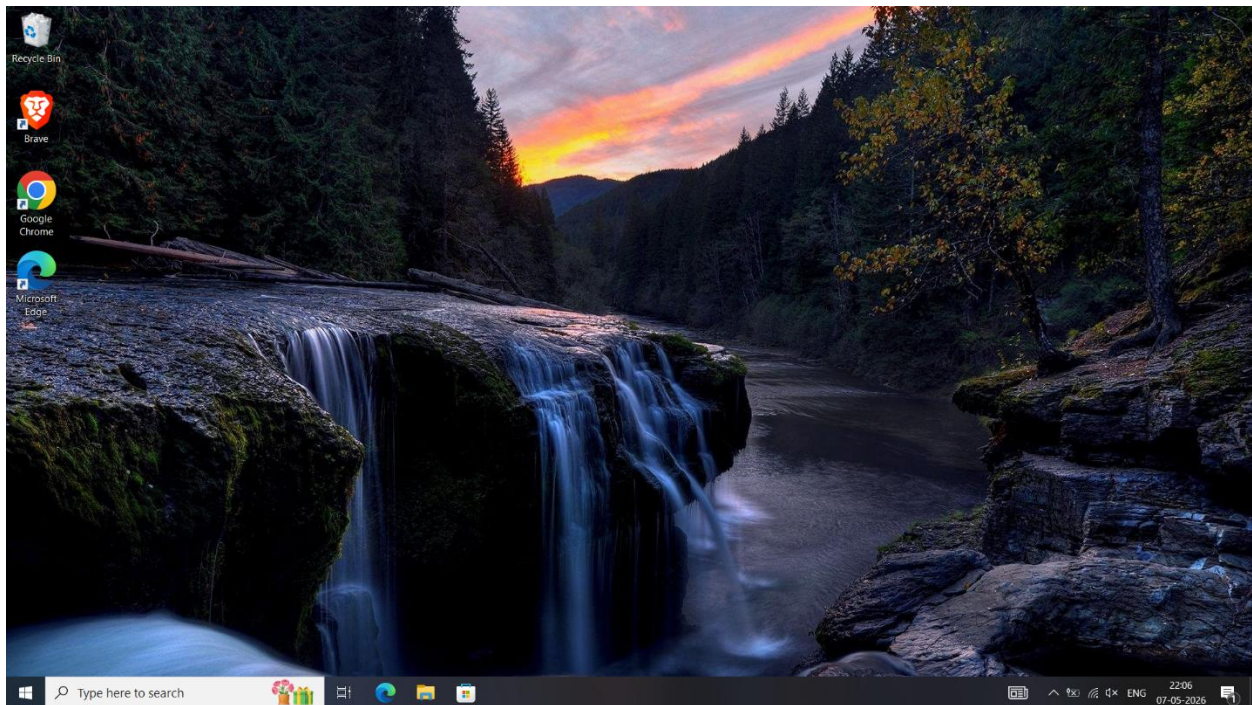
Software installation



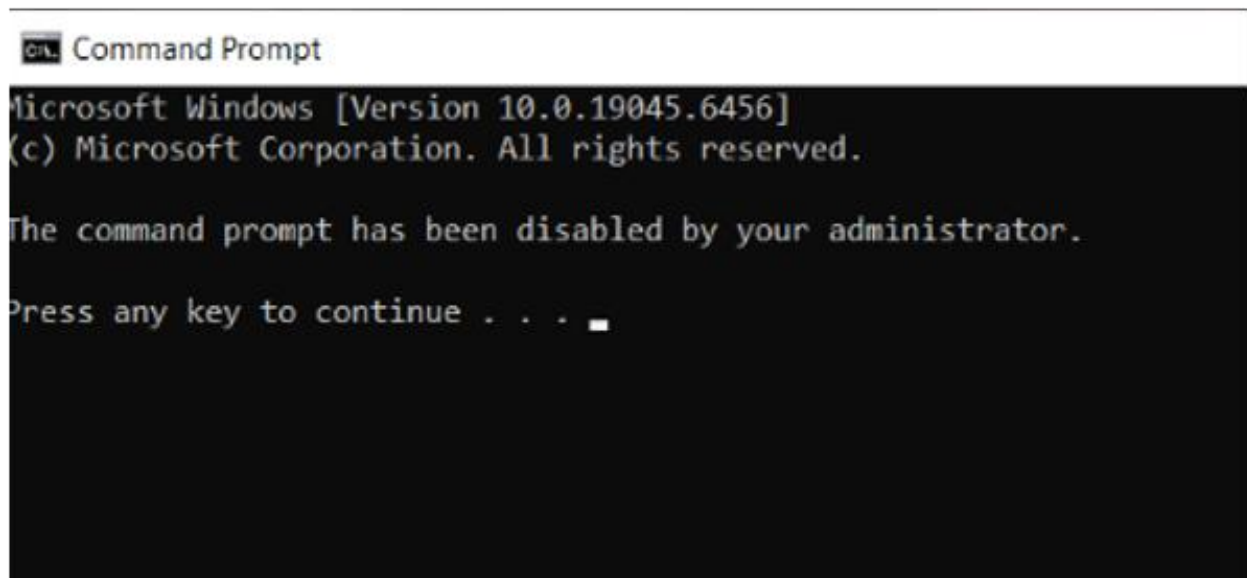
1.6 GPO Results on Client Machine

After GPO was applied, the client Dell laptop reflected all configured policies. CMD and Settings app were inaccessible for restricted users, the enforced wallpaper was applied, and USB storage was blocked. The screenshots below show the result on the client machine.

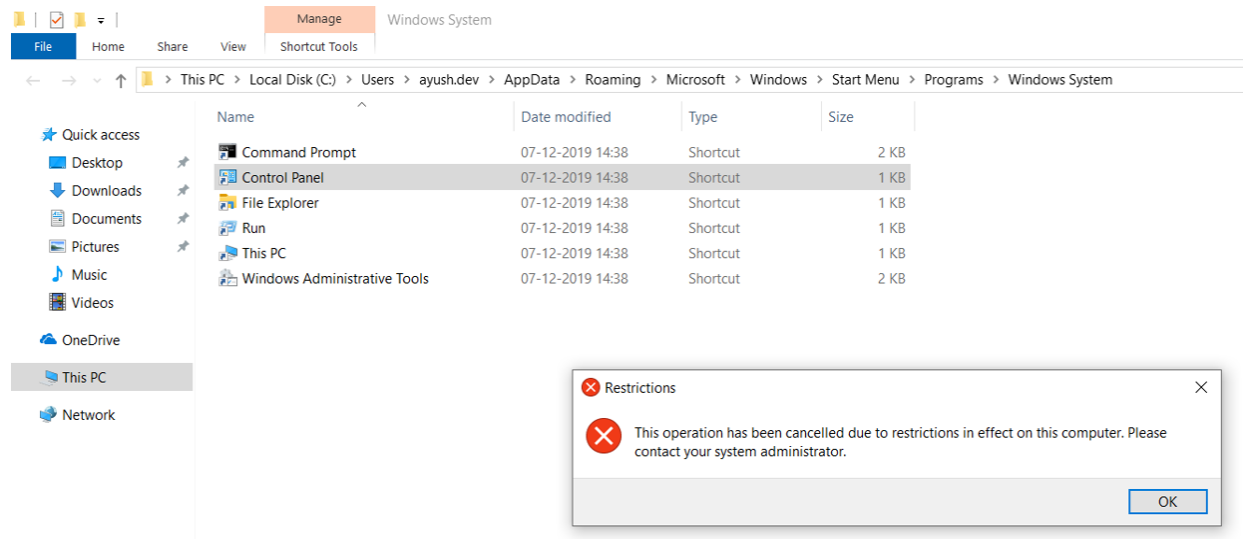
Wallpaper Enforced



Command Prompt Disabled



Control Panel Blocked



Section 2: Microsoft 365 Administration

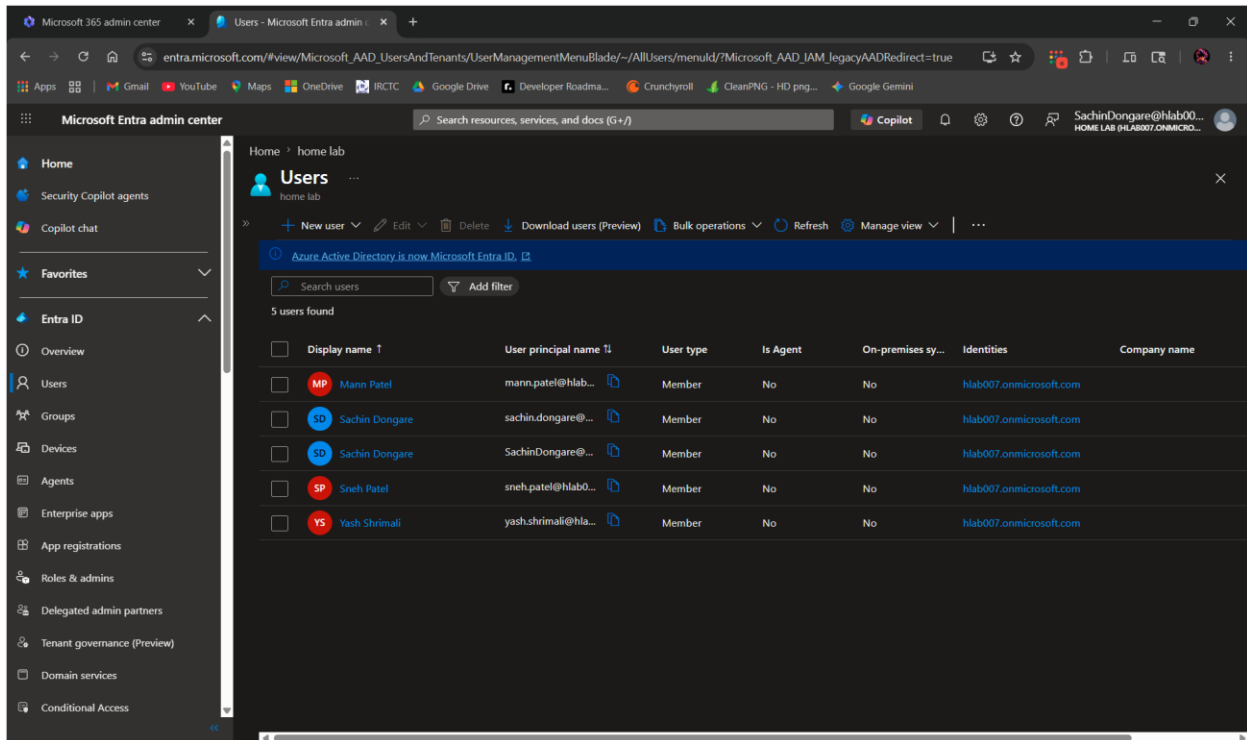
This section covers the cloud-based Microsoft 365 environment configured using a trial tenant. All administration was done through the Microsoft 365 Admin Center, Entra ID, Intune, Exchange Online, and SharePoint Online.

2.1 Entra ID — User and Group Management

Users were created in the Microsoft 365 Admin Center and assigned appropriate licenses. Groups including Security Groups, Microsoft 365 Groups, and Distribution Groups were configured to manage access and collaboration.

Users and Licenses

Multiple user accounts were created with licenses assigned. User profiles include department, job title, and contact information — reflecting how users would be set up in a real organization.

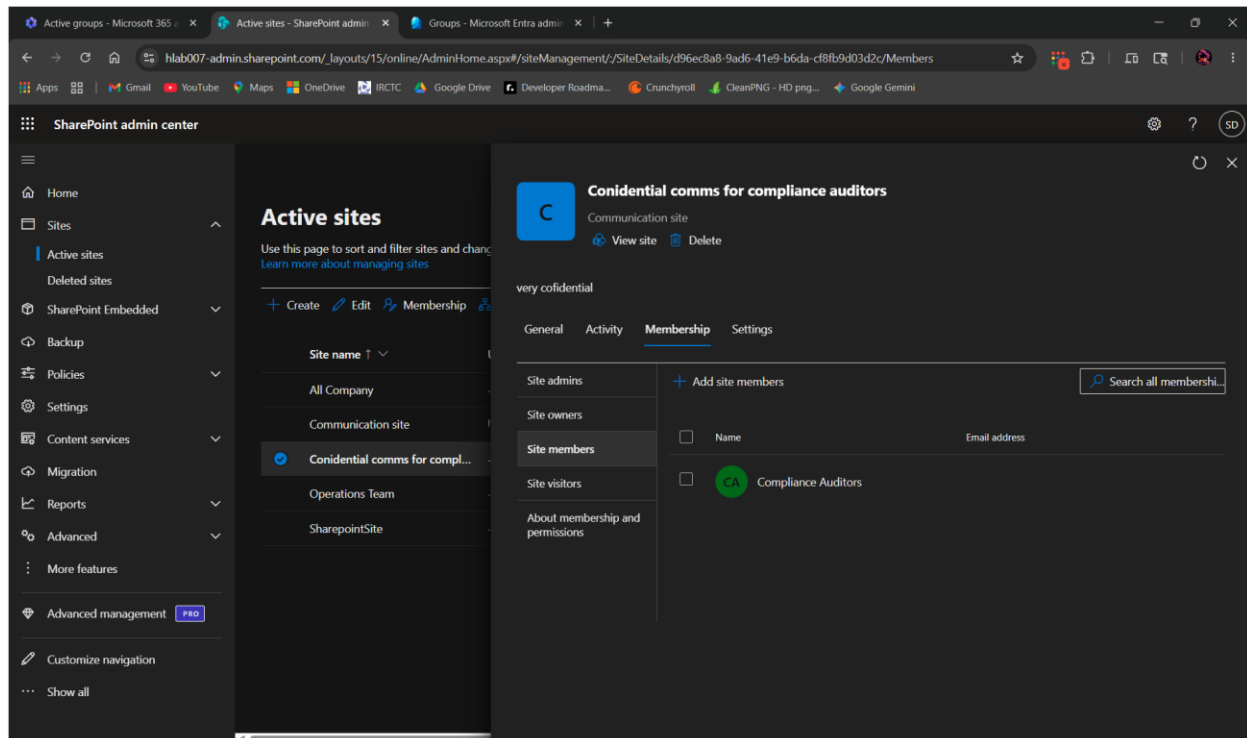
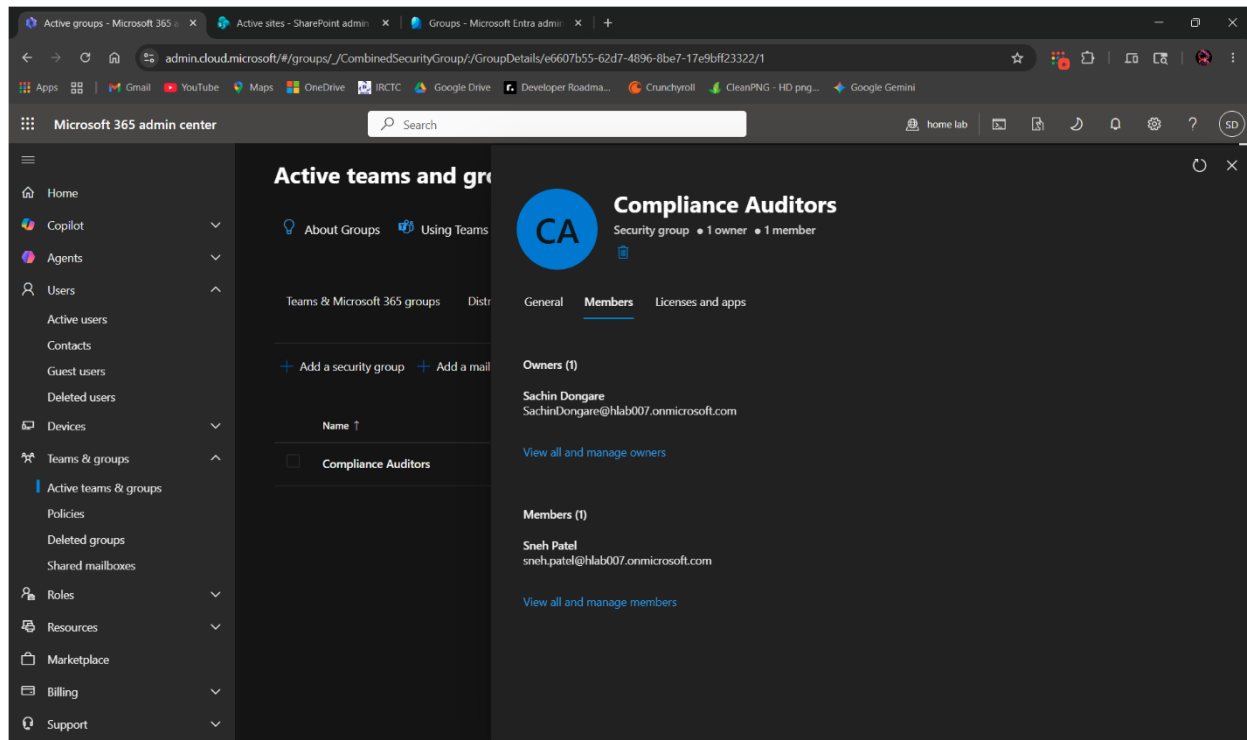


The screenshot displays the Microsoft Entra admin center interface. The left sidebar shows the navigation menu with options like Home, Security Copilot agents, Copilot chat, Favorites, and Entra ID. The main content area is titled 'Users' and shows a list of 5 users found. The table lists user details including Display name, User principal name, User type, Is Agent, On-premises sync, Identities, and Company name.

	Display name	User principal name	User type	Is Agent	On-premises sync	Identities	Company name
☐	MP Mann Patel	mann.patel@hlab...	Member	No	No	hlab007.onmicrosoft.com	
☐	S0 Sachin Dongare	sachin.dongare@...	Member	No	No	hlab007.onmicrosoft.com	
☐	S0 Sachin Dongare	SachinDongare@...	Member	No	No	hlab007.onmicrosoft.com	
☐	SP Sneh Patel	sneh.patel@hlab...	Member	No	No	hlab007.onmicrosoft.com	
☐	YS Yash Shirmali	yash.shirmali@hla...	Member	No	No	hlab007.onmicrosoft.com	

Security Groups

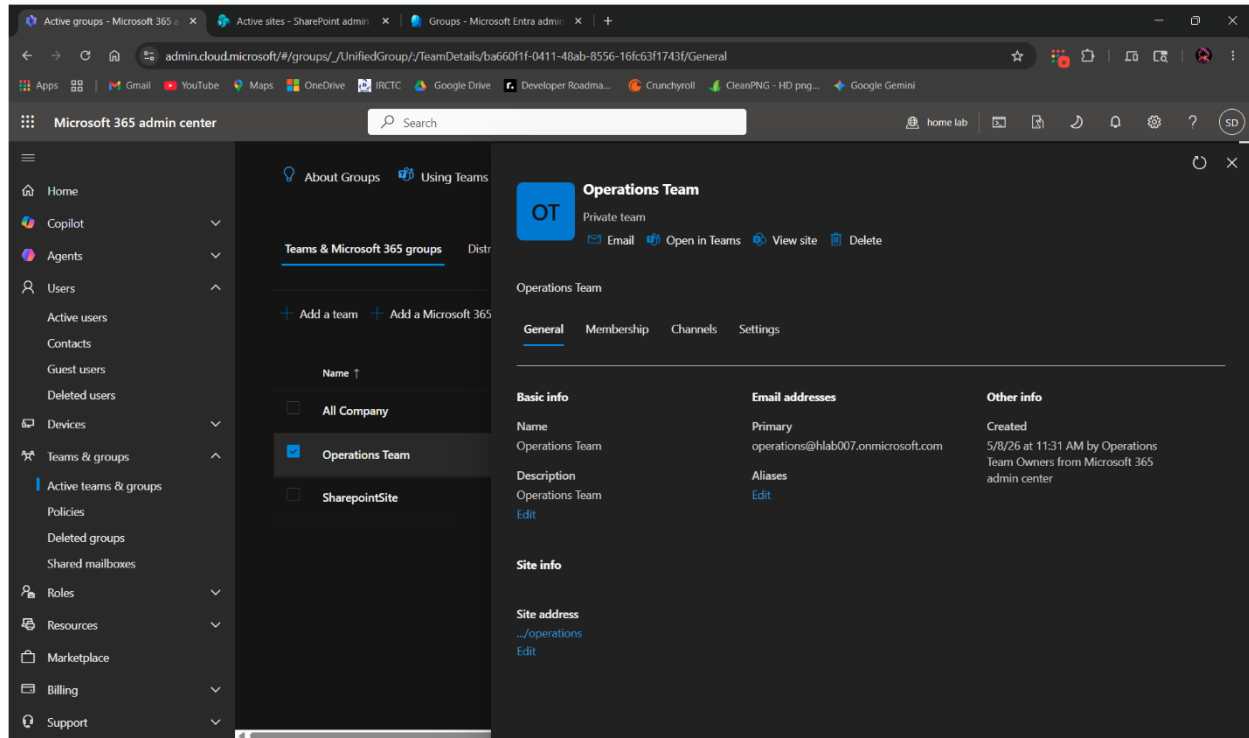
Security groups were created and users were added as members. These groups were then used to assign SharePoint site permissions, demonstrating how access control works through group membership rather than individual user assignment.



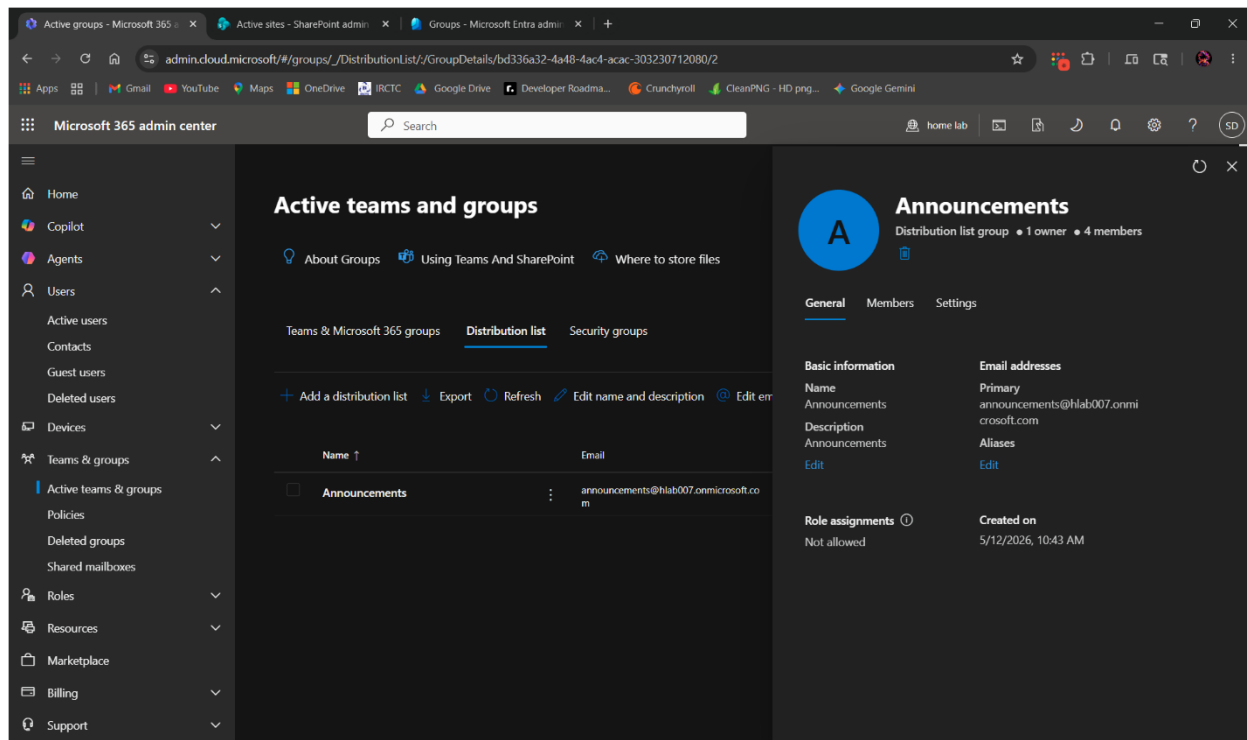
Microsoft 365 Groups and Distribution Groups

Microsoft 365 Groups were created, automatically provisioning a Teams workspace, SharePoint site, and shared mailbox. Distribution Groups were also configured for company-wide email communication.

Microsoft 365 Group



Distribution Group



Session Management

Password reset and session revocation were performed on user accounts. Revoking sessions signs the user out of all devices and apps simultaneously — a key step during offboarding or suspected account compromise.

Accounts

Shared experiences use all system accounts to authorize actions across devices

Some of your accounts require attention

Fix now

Manage your accounts

Nearby sharing

Share content with a nearby device by using Bluetooth

Off

I can share or receive content from

Everyone nearby

Save files I receive to

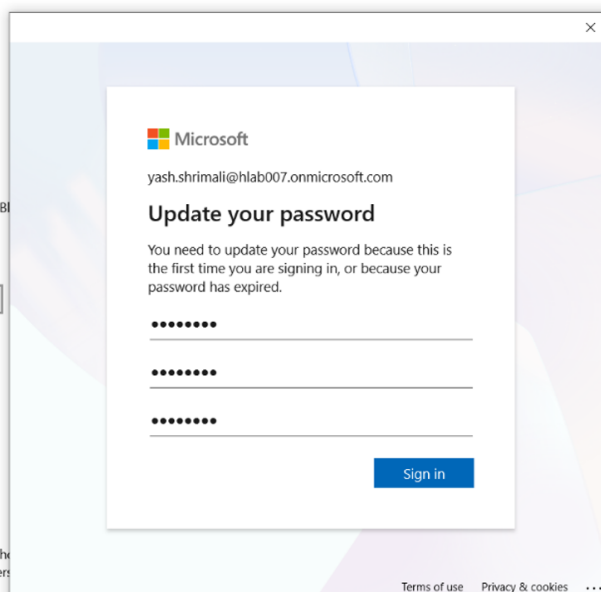
C:\Users\YashShrimali\Downloads

Change

Learn more

Share across devices

Let apps on other devices (including linked phone and message apps on this device, and vice versa)

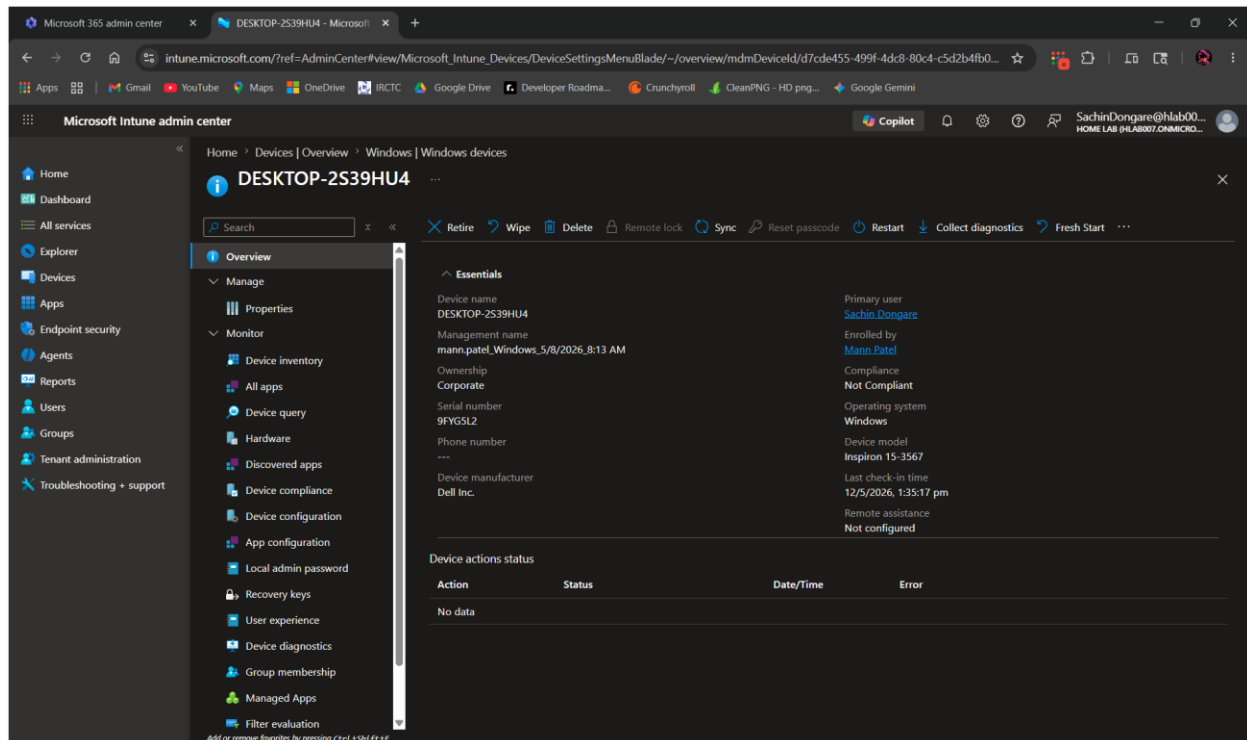


2.2 Microsoft Intune — Device Management

Devices were enrolled into Intune via Entra ID join. Compliance policies, configuration profiles, and app deployments were configured to manage and secure enrolled devices.

Device Enrollment

The test device was enrolled into Intune via Entra ID join. Once enrolled, the device appeared in the Intune portal and was visible for policy assignment and monitoring.



← Settings

Managed by home lab

Connecting to work or school allows your organization to control some things on this device, such as settings and applications.

Areas managed by home lab

home lab manages the following areas and settings. Settings marked as Dynamic might change depending on device location, time, and network configuration.

[More information about Dynamic Management](#)

Policies

- ADMX_Desktop
- System
- ADMX_RemovableStorage
- Security
- Update

Applications

- Microsoft Intune Management Extension: EnforcementCompleted

Connection info

Management Server Address:

<https://r.manage.microsoft.com/devicegatewayproxy/cimhandler.aspx>

Exchange ID:

A6027B4C27CD5A40D028E47A74CFB381

Device sync status

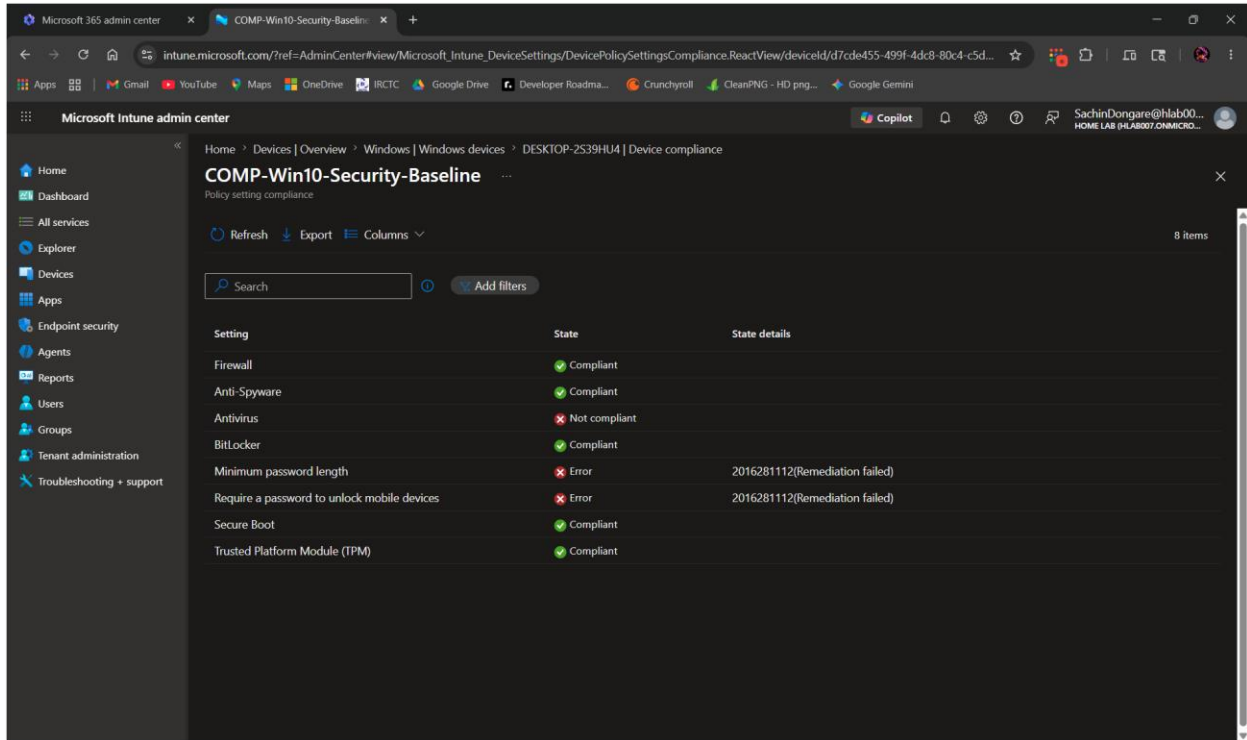
Syncing home security policies, network profiles, and managed



Compliance Policy

A device compliance policy was configured with the following requirements:

- BitLocker encryption must be enabled
- Minimum Windows version enforced
- Antivirus must be active and reporting
- Minimum password length requirement



Microsoft Intune admin center

COMP-Win10-Security-Baseline

Home > Devices > Overview > Windows > Windows devices > DESKTOP-2S39HU4 | Device compliance

COMP-Win10-Security-Baseline

Policy setting compliance

Refresh Export Columns

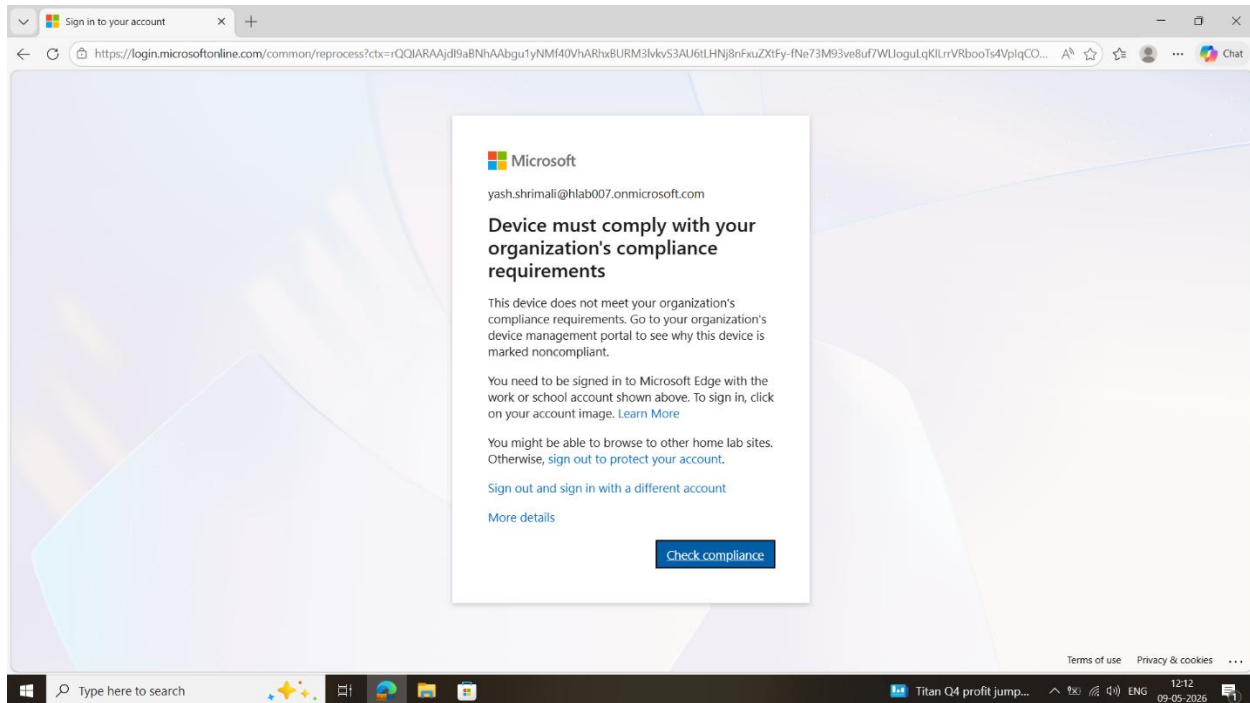
8 Items

Search Add filters

Setting	State	State details
Firewall	Compliant	
Anti-Spyware	Compliant	
Antivirus	Not compliant	
BitLocker	Compliant	
Minimum password length	Error	2016281112(Remediation failed)
Require a password to unlock mobile devices	Error	2016281112(Remediation failed)
Secure Boot	Compliant	
Trusted Platform Module (TPM)	Compliant	

Non-Compliant Device — Email Block

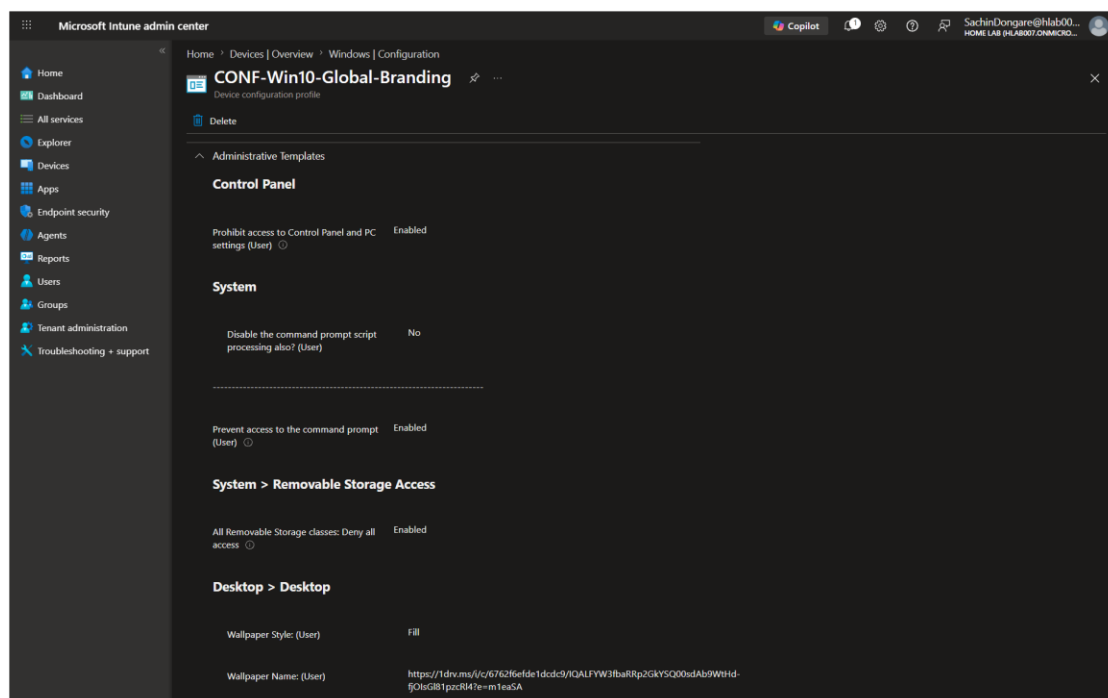
A Conditional Access policy was configured to block non-compliant devices from accessing company email. When the test device was marked non-compliant, access to the mailbox was blocked — demonstrating end-to-end enforcement of device compliance.



Configuration Profiles

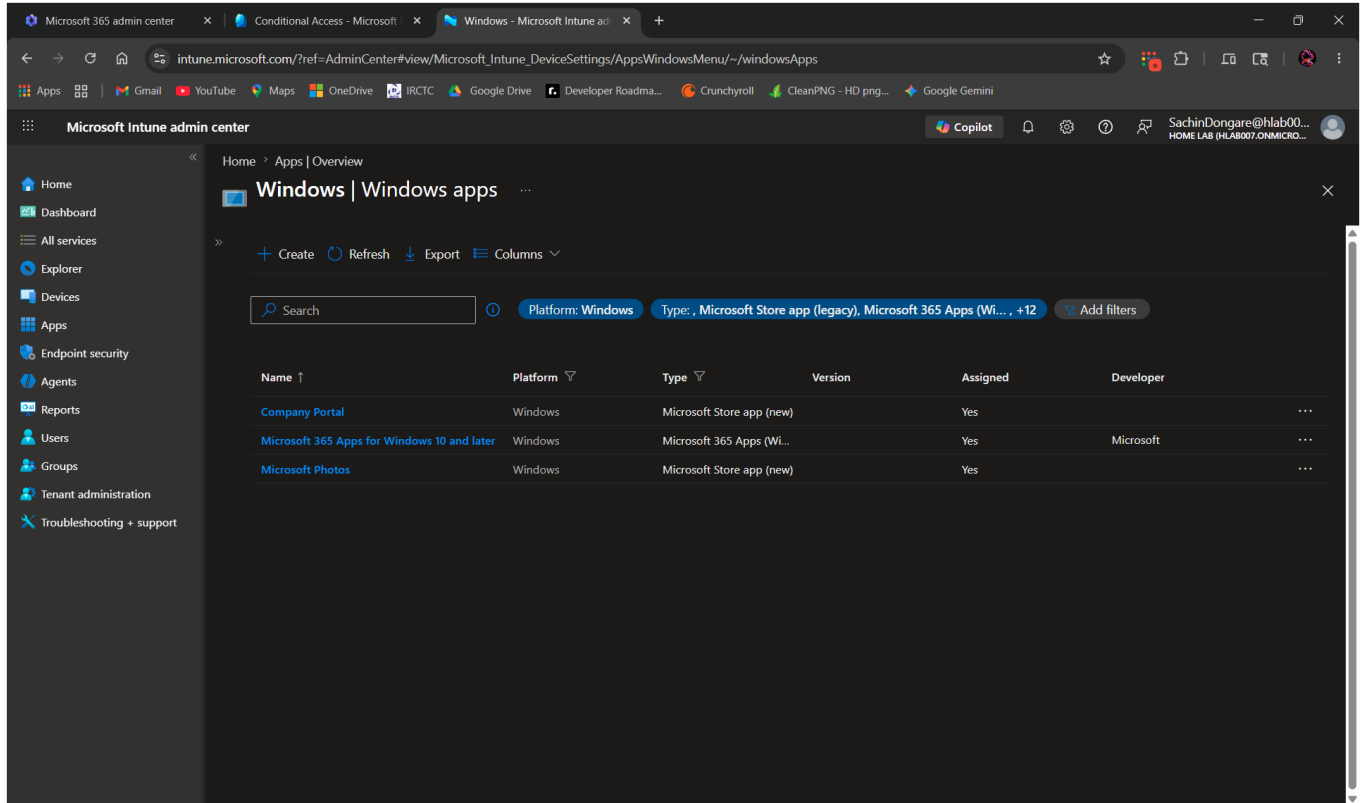
Configuration profiles were created and deployed to enforce the following settings on managed devices:

- USB storage blocked to prevent unauthorized data transfer
- Command Prompt (CMD) access restricted
- Control Panel access blocked for standard users



App Deployment

Microsoft 365 Apps and Company Portal were deployed to enrolled devices via Intune app deployment. Apps were assigned to user groups and installed automatically without any manual intervention on the device.



The screenshot displays the Microsoft Intune admin center interface. The left sidebar contains navigation options: Home, Dashboard, All services, Explorer, Devices, Apps, Endpoint security, Agents, Reports, Users, Groups, Tenant administration, and Troubleshooting + support. The main content area is titled 'Windows | Windows apps' and shows a list of deployed applications. The list includes 'Company Portal', 'Microsoft 365 Apps for Windows 10 and later', and 'Microsoft Photos'. Each entry shows its platform (Windows), type (Microsoft Store app), version, assignment status (Yes), and developer (Microsoft).

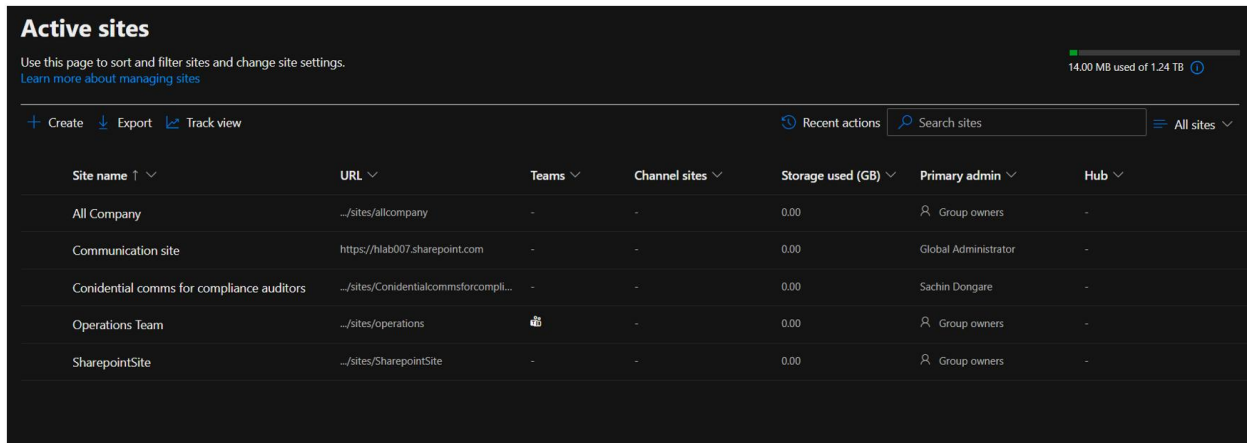
Name	Platform	Type	Version	Assigned	Developer
Company Portal	Windows	Microsoft Store app (new)		Yes	
Microsoft 365 Apps for Windows 10 and later	Windows	Microsoft 365 Apps (Wi...		Yes	Microsoft
Microsoft Photos	Windows	Microsoft Store app (new)		Yes	

2.3 SharePoint Online

SharePoint sites were created and configured to demonstrate site management, permissions, and access control in a Microsoft 365 environment.

Team Site and Communication Site

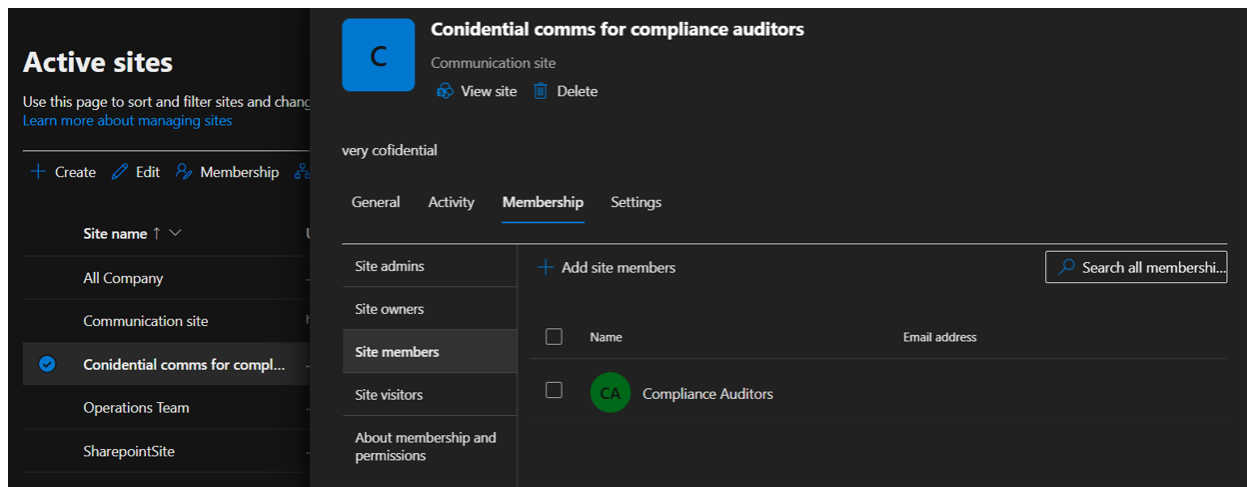
Both a Team Site and a Communication Site were created. The Team Site was linked to a Microsoft 365 Group, automatically provisioning a shared workspace. The Communication Site was created as a standalone portal for broadcasting information to a wider audience.



Site name ↑ ▾	URL ▾	Teams ▾	Channel sites ▾	Storage used (GB) ▾	Primary admin ▾	Hub ▾
All Company	.../sites/allcompany	-	-	0.00	Group owners	-
Communication site	https://hlab007.sharepoint.com	-	-	0.00	Global Administrator	-
Confidential comms for compliance auditors	.../sites/Confidentialcommsforcompli...	-	-	0.00	Sachin Dongare	-
Operations Team	.../sites/operations		-	0.00	Group owners	-
SharepointSite	.../sites/SharepointSite	-	-	0.00	Group owners	-

Site Permissions

Permissions were configured on the Communication Site by assigning the previously created Security Group as Members. This demonstrates how group-based access control works in SharePoint — access is managed through group membership rather than individual users.



Confidential comms for compliance auditors
Communication site
[View site](#) [Delete](#)

very confidential

General Activity **Membership** Settings

Site admins

Site owners

Site members

Site visitors

About membership and permissions

+ Add site members

Search all membershi...

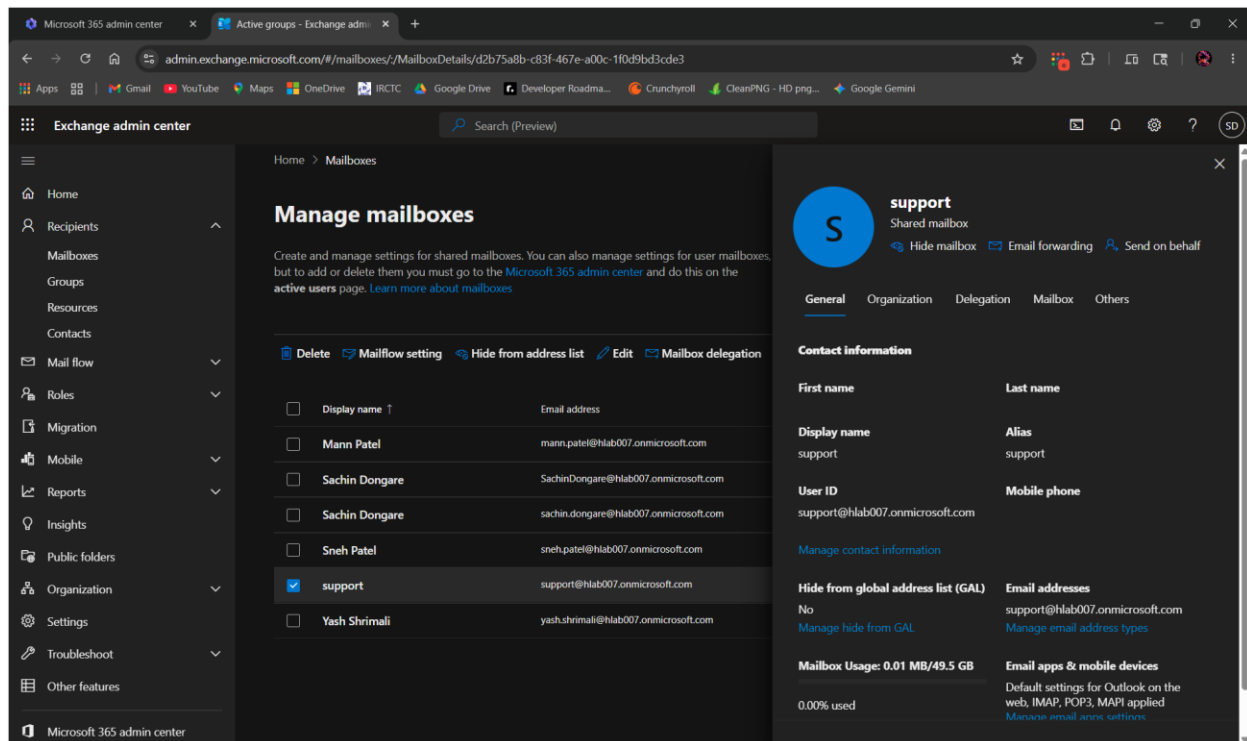
☐	Name	Email address
☐	CA	Compliance Auditors

2.4 Exchange Online

Exchange Online was configured to demonstrate mailbox management and mail flow rule implementation.

Shared Mailbox

A shared mailbox was created and team members were added with Full Access and Send As permissions. This allows multiple users to monitor and reply from a single email address — commonly used for support or team inboxes.

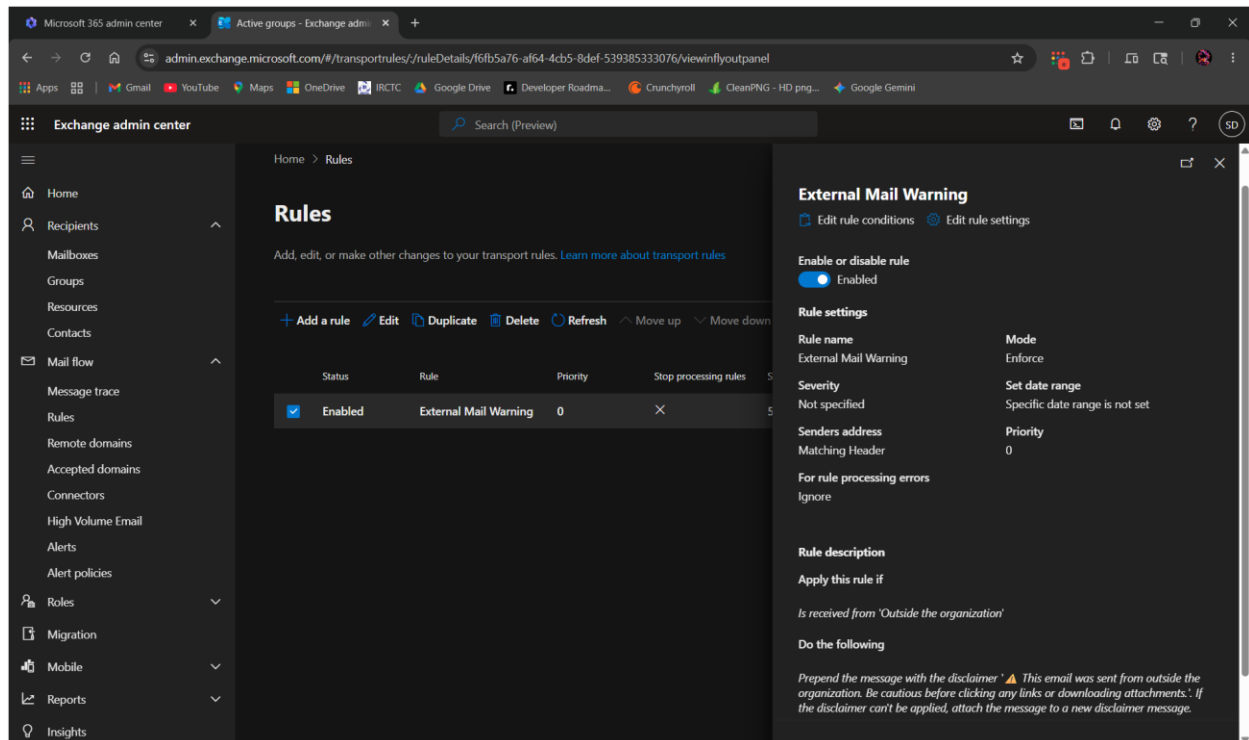


Mail Flow Rule — External Email Warning

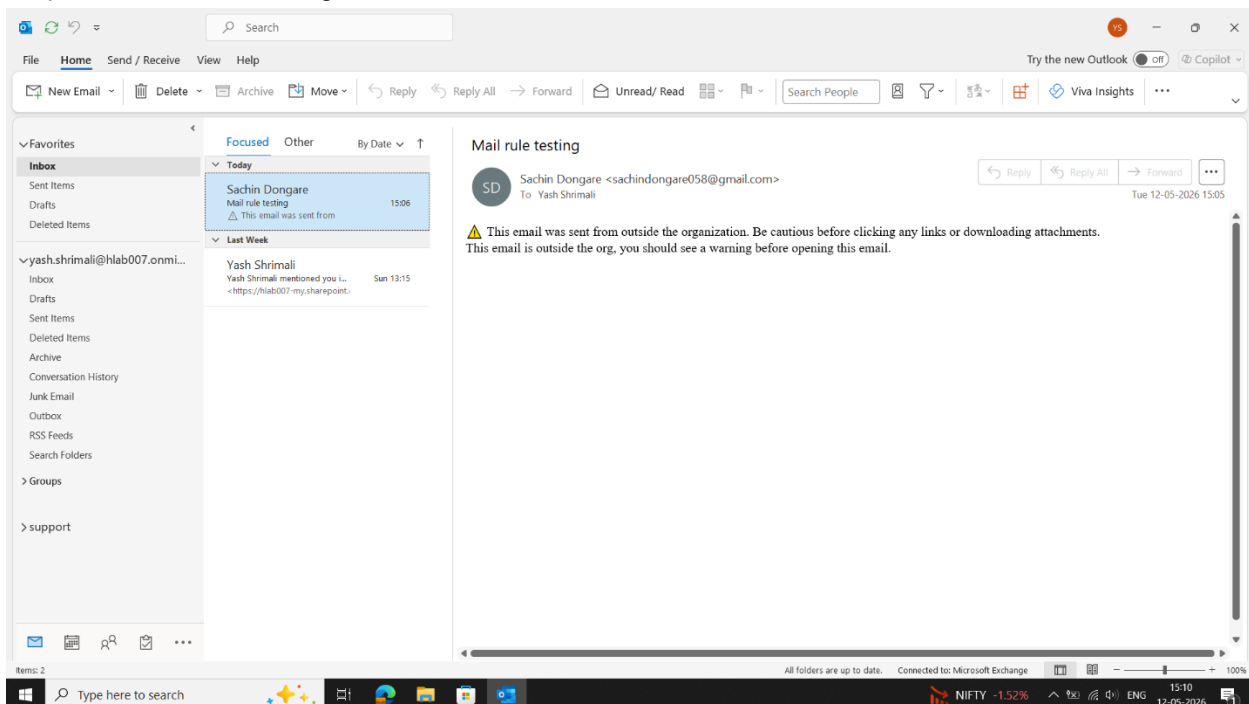
A mail flow rule was configured to automatically prepend a warning banner to any email received from outside the organization. This alerts employees to be cautious with external emails and helps reduce the risk of phishing attacks.

- Condition: Sender is located outside the organization
- Action: Prepend disclaimer warning banner to the email

The rule was tested by sending an email from a personal Gmail account to the Microsoft 365 mailbox. The warning banner appeared at the top of the received email confirming the rule worked correctly.



Prepend disclaimer warning on user's mail.



Conclusion

This home lab covered the core areas of enterprise IT administration — from on-premise Active Directory and Group Policy to cloud-based Microsoft 365 services including Entra ID, Intune, Exchange Online, and SharePoint Online.

Every configuration in this report was implemented hands-on, tested, and verified on real or virtualized hardware. The lab was built to develop practical skills that translate directly to IT support and system administration roles.